Cox Regression as Regression

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Introduction

Cox proportional-hazards regression models time-to-event outcomes with right censoring and is the de facto standard regression tool for survival analysis. This page gives a regression-modelling-focused summary; the Survival Analysis section contains a more detailed treatment with diagnostics, residuals, and stratification. The headline insight is that Cox is a regression in the same family as GLMs — coefficients, standard errors, confidence intervals, and likelihood-ratio tests all behave familiarly — but the response is a `Surv` object encoding both event time and censoring status.

Prerequisites

A working understanding of survival analysis, the hazard function, censoring, and the partial-likelihood approach to inference.

Theory

The Cox model writes the hazard as

$$h(t \mid X) = h_0(t) \exp(X^\top \beta),$$

with the baseline hazard $h_0(t)$ left unspecified and the partial likelihood providing $\hat{\beta}$ without estimating it. Exponentiated coefficients are hazard ratios; a $\hat{\beta}_j = 0.02$ for age corresponds to a 2 % multiplicative increase in hazard per year of age, $\exp(0.02) \approx 1.02$.

The proportional-hazards assumption — that hazard ratios between any two covariate patterns are constant over time — is the model's defining restriction. When it fails, options include stratification, time-varying coefficients via `tt()` or `survSplit()`, or switching to AFT or RMST-based summaries.

Assumptions

Proportional hazards across covariate levels (test via Schoenfeld residuals), independent non-informative censoring, no unmodelled time-varying covariate effects, and adequate events per variable for stable estimation (rule of thumb: 10–15 events per regression coefficient).

R Implementation

```r
library(survival); library(broom)

data(lung)
fit <- coxph(Surv(time, status) ~ age + sex + ph.ecog, data = lung)
tidy(fit, conf.int = TRUE, exponentiate = TRUE)

# PH check
cox.zph(fit)
```

Output & Results

`tidy()` from `broom` returns a tibble of hazard ratios, confidence intervals, and Wald p-values. `summary(fit)` adds the likelihood-ratio test, score test, and Wald test for the joint null. `cox.zph()` reports the proportional-hazards diagnostic for each covariate plus a global test.

Interpretation

A reporting sentence: "A multivariable Cox model adjusted for age, sex, and ECOG performance status estimated that each additional year of age increased the hazard of death by 2 % (HR 1.02, 95 % CI 1.00 to 1.04, $p = 0.04$); female sex was associated with a 40 % lower hazard (HR 0.60, 95 % CI 0.42 to 0.85, $p = 0.004$). The proportional-hazards assumption was supported (global Schoenfeld $p = 0.55$)." Always pair coefficient reporting with PH diagnostics.

Practical Tips

- Always run `cox.zph()` after fitting; reporting Cox HRs without PH diagnostics is no longer acceptable in clinical-trial publications.
- Use Efron's tie-handling method (`ties = "efron"`) for tied event times; the default in older software is Breslow's, which is less accurate.
- Report HRs with 95 % CIs alongside p-values; a CI that includes 1.0 is more informative than a marginal p-value alone.
- For time-varying covariates, use counting-process format (`Surv(tstart, tstop, event)`) and cluster standard errors on subject id.
- For continuous covariates with non-linear effects, add restricted cubic splines (`rms::rcs(x, 4)`) rather than dichotomising.
- See the Survival Analysis section's dedicated Cox tutorial for extended coverage of residuals, stratification, and influence diagnostics.

R Packages Used

`survival` for `coxph()`, `Surv()`, `cox.zph()`; `broom` for `tidy()` and `glance()` summaries; `survminer` for ggplot-style diagnostics; `rms::cph()` for an alternative interface with built-in spline and validation tools.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls

- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI